University of Kelaniya

Department of Finance

BFIN 22623 / BBNF 22223 / BINF 22223 / BBFE 22333

Financial Data Science Group Assignment

Applied Financial Data Analytics and Decision Reporting Using Python

---

Total Marks: 30

Group Size: 10 students

Due Date: 31.10.2026

Objective

This assignment is an applied financial analytics project. Each group must build a reproducible Python workflow, analyse real market data, interpret the outputs for a decision maker, and prepare a professional report. Marks are awarded for working code, justified data decisions, accurate interpretation, and clear communication rather than for displaying outputs alone.

Real World Scenario

Your group works as a Junior Financial Analytics Team in an investment research firm. The Investment Committee requires a market-monitoring briefing that compares selected assets, identifies return and risk patterns, highlights unusual observations, and evaluates possible diversification relationships. The Committee will use your briefing to decide which assets require closer monitoring and further investigation.

Select four listed shares or other approved financial assets and one relevant market index as a benchmark. Use at least 24 months of daily data. Sri Lankan market data are encouraged when a reliable source is available; otherwise, use a lecturer-approved international market dataset. Data must not be fabricated or manually altered.

Scope of the Assignment

- Topic 01 - Financial Data Science, financial analytics, real-world business questions, data sources, and data limitations.
- Topic 02 - Python packages, reusable functions, file handling, data formats, pandas DataFrames, data cleaning, and reproducible workflows.
- Topic 03 - Financial returns, descriptive statistics, covariance, correlation, and decision-oriented visualization.

Portfolio optimization, regression, machine-learning prediction, and time-series forecasting are outside the scope of this assignment.

Tasks and Instructions

Task 1: Decision Brief and Data Plan (3 Marks)

Prepare a short decision brief before writing the analysis code. It must establish who will use the analysis, what decision the analysis supports, and what evidence will be produced.

1. Identify the four assets and benchmark, and justify their relevance to the decision scenario.
2. State three practical questions covering performance, risk, and co-movement or diversification.

3. Record the data source, observation period, variables, units, frequency, and known limitations in a concise data dictionary.
4. Explain how the proposed workflow applies Financial Data Science to a real financial decision.

Deliverables

- A 250-350 word decision brief and a data dictionary presented in the opening section of the notebook.

Task 2: Reproducible Data Pipeline and Coding (5 Marks)

Build a Python pipeline that another analyst can run from beginning to end without manually changing the source data. Use relevant packages such as pandas, NumPy, matplotlib, pathlib, openpyxl, and yfinance where appropriate.

- Create a configuration section for tickers or asset names, dates, folders, and output filenames.
- Acquire data through an API or import an approved CSV or Excel file, then preserve an unchanged raw copy.
- Display the first and last observations, dimensions, data types, date range, and missing-value counts.
- Use relative paths with pathlib; hard-coded personal computer paths are not accepted.
- Create a reusable finance_tools.py module containing at least three meaningful functions, such as load_data(), validate_data(), calculate_returns(), or save_outputs().
- Include clear comments, docstrings, and basic error handling for missing files, invalid dates, or unavailable tickers.

Deliverables

- A fully executed notebook, the finance_tools.py module, the unchanged raw dataset, and a data-audit table.

Task 3: Data Quality and Feature Engineering (3 Marks)

Clean and transform the data entirely through Python. Explain each material decision in a Markdown cell immediately before or after the relevant code.

- Convert dates and numerical fields, sort records, remove duplicates, and select the appropriate price field.
- Identify missing values and justify the treatment used. Do not silently replace or delete observations.
- Investigate extreme price changes and decide whether they represent valid market movements, corporate actions, or data errors.
- Calculate daily simple returns and a normalized price index beginning at 100 for each asset and the benchmark.
- Create automated checks using assertions or equivalent tests for duplicate dates, date order, numerical prices, missing analytical values, and successful output-file creation.

$$Return_t = (Price_t / Price_{t-1}) - 1$$

Deliverables

- clean_financial_data.csv, data_quality_log.csv, visible validation results, and written justification for the main cleaning decisions.

Task 4: Statistical Analysis and Financial Interpretation (3 Marks)

Produce an analyst-ready summary table for every asset and the benchmark. The table must contain the following measures calculated from the daily return series.

Statistic	Required	Statistic	Required
Number of observations	Yes	Variance	Yes
Mean daily return	Yes	Standard deviation	Yes
Minimum and maximum	Yes	Skewness	Yes
Total period return	Yes	Kurtosis	Yes

Calculate covariance and correlation matrices for the assets and benchmark. After every main table or matrix, use the following interpretation sequence: Observation - numerical evidence - financial implication - limitation. Identify the strongest and weakest performance, highest and lowest volatility, unusual distribution characteristics, relationships with the benchmark, and potentially useful diversification pairs. Provide at least three evidence-based findings that an Investment Committee could act upon.

Deliverables

- summary_statistics.csv, covariance_matrix.csv, correlation_matrix.csv, and concise financial interpretation linked to numerical evidence.

Task 5: Decision Dashboard and Visual Interpretation (3 Marks)

Create a compact set of decision-oriented figures that could be used in an Investment Committee briefing. The figures must include:

1. A normalized price index chart comparing all assets and the benchmark from a common starting value of 100.
2. Return-distribution histograms or comparable distribution plots.
3. A risk-return chart using mean return and standard deviation, with each point labelled.
4. A correlation heatmap covering the selected assets and benchmark.

Every figure must contain a decision-focused title, axis labels, readable legend, source or period note, and a two- or three-sentence interpretation. The interpretation must state what happened, the evidence visible in the figure, and why it matters. Decorative graphs that do not answer an analytical question will not receive marks.

Deliverables

- At least four exported PNG figures and their interpretations in both the notebook and report.

Task 6: Investment Committee Report (3 Marks)

Prepare a professional 1,500-2,000 word report for the Investment Committee. The report must stand on its own and must not read like a list of Python outputs.

1. Executive Summary of 200-250 words stating the decision context, three principal findings, and the main recommendation.
2. Data and Method describing the source, period, variables, cleaning decisions, return calculation, and limitations.

3. Findings and Interpretation integrating the most relevant tables and figures with evidence-based discussion.
4. Recommendation identifying which assets require closer monitoring or further analysis and explaining why. Do not present the work as personal investment advice.
5. Conclusion, limitations, references, and an appendix containing the data dictionary and contribution statement.

Deliverables

- A professionally formatted PDF report of 1,500-2,000 words, excluding appendices and references.

Task 7: Group Presentation and Question Defence (10 Marks)

Deliver an 8-minute Investment Committee briefing followed by 2 minutes of questions. The presentation must communicate the group's decision-relevant evidence clearly; it must not simply repeat the report or read text from the slides.

1. Prepare 6-8 slides covering: the decision question and selected assets; data source and period; the reproducible workflow and key data-quality decisions; the most important statistical findings; two or three decision-oriented visualizations; recommendation; limitations; and references.

2. Support every conclusion with a statistic, table, or figure from the submitted analysis. Values, labels, periods, and recommendations must be consistent across the notebook, report, and slides.

3. Every group member must have a clearly identified role in presenting or answering questions. Record each member's presentation role in the contribution statement.

4. Keep within the allocated time and be prepared to defend the asset selection, data source, cleaning decisions, return calculations, statistical interpretation, recommendations, and limitations.

Presentation Marking

- Analytical content and use of evidence - 3 marks
- Financial interpretation and recommendation - 3 marks
- Slide quality and professional delivery - 1 mark
- Team participation and time management - 1 mark
- Responses to questions and ability to defend decisions - 2 marks

Deliverables

- Group_XX_Briefing.pdf, an 8-minute live group presentation, a 2-minute question session, and presentation roles recorded in the signed contribution statement.

Coding and Evidence Requirements

- The notebook must run from the first cell to the last cell without errors and all outputs must be visible.
- Screenshots of code are not accepted. Submit editable Python code and the files generated by that code.
- Every major table and figure must be followed by interpretation; marks are not awarded for unexplained output.
- Manual editing of raw or cleaned data is not permitted. The submitted workflow must reproduce the final outputs.
- Use meaningful names, comments, docstrings, relative paths, and appropriate exception handling.

Submission Guidelines

1. One ZIP folder using the prescribed project structure and naming convention.

2. Jupyter Notebook (.ipynb) with executed code, outputs, explanations, and interpretations.
3. Python module (.py), raw and cleaned datasets, audit log, analytical output files, and exported figures.
4. Investment Committee Report (PDF), 1,500-2,000 words excluding appendices and references.
5. Briefing slides in PDF format and a signed contribution statement showing each member's presentation or question-defence role. Every member must participate.

Required Folder and File Structure

```
Group_XX_Financial_Data_Science/
    data/raw/raw_market_data.csv
    data/processed/clean_financial_data.csv
    src/finance_tools.py
    notebook/Group_XX_Market_Analysis.ipynb
    outputs/data_quality_log.csv
    outputs/summary_statistics.csv
    outputs/covariance_matrix.csv
    outputs/correlation_matrix.csv
    outputs/figures/
    report/Group_XX_Committee_Report.pdf
    presentation/Group_XX_Briefing.pdf
    contribution/Group_XX_Contribution_Statement.pdf
```

Evaluation Criteria

Assessment Component	Marks
Decision Brief and Data Plan	3
Reproducible Data Pipeline and Coding	5
Data Quality and Feature Engineering	3
Statistical Analysis and Financial Interpretation	3
Decision Dashboard and Visual Interpretation	3
Investment Committee Report	3
Group Presentation and Question Defence	10
Total	**30**

General Requirements

- Use real, traceable data and record the exact source, retrieval date, and observation period.
- Ensure consistency between the notebook, exported data, reported statistics, figures, and conclusions.
- Use consistent units, appropriate decimal places, and clear table and figure labels.
- Cite datasets, documentation, external code fragments, and other sources used.
- State important data and method limitations; unsupported claims will reduce interpretation marks.

References and Recommended Documentation

- pandas Documentation - https://pandas.pydata.org/docs/
- NumPy Documentation - https://numpy.org/doc/

- Matplotlib Documentation - https://matplotlib.org/stable/
- yfinance Documentation - https://ranaroussi.github.io/yfinance/

Note: Plagiarism, unacknowledged submission of copied code, falsified contribution statements, or other academic misconduct may result in zero marks, subject to University regulations. All code and explanations must represent the group's own work, and any permitted AI-assisted content must be declared according to the lecturer's instructions.